

MAJOR PROJECT REGISTRATION FORM

Project Category:	☒ Research oriented	☐ Application oriented	Date:	Batch ID: A08
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Mental Health Detection Using EEG Signals

ABSTRACT

Mental health problems such as depression, anxiety, and stress are becoming common among people of different age groups. These conditions can affect a person's emotions, behavior, studies, work, relationships, and overall quality of life. Early identification of mental health problems is important because it can help individuals receive proper guidance and support at an early stage. However, traditional methods of detecting these conditions mainly depend on questionnaires, interviews, and observations. These methods may sometimes be subjective and depend on the person's ability to describe their feelings accurately.

Electroencephalography (EEG) is a non-invasive technique used to record the electrical activity of the brain through electrodes placed on the scalp. EEG signals contain information about brain activity and may show changes related to different mental states. Therefore, EEG can be used as a useful source of data for studying and detecting mental health conditions.

This project focuses on studying how EEG signals can be used for the detection of mental health conditions such as depression, anxiety, and stress. The EEG data is first processed to remove unwanted noise, artifacts, and disturbances. Important features are then extracted from the processed signals using time-domain, frequency-domain, and time-frequency-domain techniques. These features are used as input for machine learning and deep learning models.

Different algorithms such as Support Vector Machine (SVM), Random Forest, Logistic Regression, and Convolutional Neural Network (CNN) can be studied and compared for classifying mental health conditions. The performance of these models can be evaluated using measures such as accuracy, precision, recall, F1-score, and confusion matrix. The main aim of this project is to understand the relationship between EEG brain signals and mental health conditions and to identify suitable machine learning or deep learning techniques for reliable classification. The project also studies challenges such as variations in EEG signals between individuals, limited datasets, noise in EEG recordings, and the difficulty of obtaining high-quality data. The proposed system is intended to support early screening and research in mental health assessment. It is not intended to replace professional medical diagnosis.

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References (Category specific)

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- 2) Liu, Y., Pu, C., Xia, S., Deng, D., Wang, X., & Li, M. (2022). Machine learning approaches for diagnosing depression using EEG: A review. *Translational Neuroscience*, 13(1), 224–235. <https://doi.org/10.1515/tnsci-2022-0234>
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